CS228 Logic for Computer Science
Exercise Problem Set 6

Note on countermodels: To disprove a claim such as "φ is valid" or "$\Gamma \models \varphi$", we exhibit a *countermodel*: a structure M (together with an environment α, if φ has free variables) in which the claim fails. For "φ is valid" this means $M, \alpha \not\models \varphi$. For "$\Gamma \models \varphi$" it means $M, \alpha \models \gamma$ for every $\gamma \in \Gamma$, but $M, \alpha \not\models \varphi$. For a sentence we simply write $M \not\models \varphi$.

Part 1

1. Translating between English and first-order logic.

   (a) Translate the following sentences into first-order formulas over the vocabulary $\mathcal{V}$ with the constant i (Ismail), the unary predicates $S(x)$: "x is a student", $P(x)$: "x is a sport" and $L(x)$: "x is a lecture", and the binary predicates $A(x, y)$: "x likes y" and $B(x, y)$: "x attended y".

      i. Ismail likes every sport.

      ii. No student attended every lecture.

      iii. Every student attended at least two lectures.

      iv. Ismail likes exactly one sport.

   (b) Let $f(x)$ be a function symbol denoting the father of x. Translate: nobody is their own grandfather.

   (c) Over the vocabulary $\mathcal{V}$ of part (a), say in plain English what $\exists x(S(x) \wedge \forall y(A(x, y) \rightarrow P(y)))$ means.

   (d) Let $L(x, y)$ mean "x loves y", over the universe of all people. Translate $\forall x \exists y L(x, y)$ and $\exists y \forall x L(x, y)$ into English. Does either of them imply the other? Justify your answer.

2. Let φ be $\exists x(P(y, z) \wedge (\forall y(\neg Q(y, x) \vee P(y, z))))$, where P and Q are predicate symbols with two arguments.

   (a) Draw the parse tree of φ.

   (b) Identify all bound and free variable leaves in φ.

   (c) Is there a variable in φ which has free and bound occurrences?

   (d) Consider the terms w (w is a variable), $f(x)$ and $g(y, z)$, where f and g are function symbols with arity 1 and 2, respectively.

      i. Compute $\varphi[w/x]$, $\varphi[w/y]$, $\varphi[f(x)/y]$ and $\varphi[g(y, z)/z]$.

      ii. Which of w, $f(x)$ and $g(y, z)$ are free for x in φ?

      iii. Which of w, $f(x)$ and $g(y, z)$ are free for y in φ?

   (e) What is the scope of $\exists x$ in φ?

   (f) Suppose that we change φ to $\exists x(P(y, z) \wedge (\forall x(\neg Q(x, x) \vee P(x, z))))$. What is the scope of $\exists x$ now?

3. Let M be a model with the universe $U^M = \mathbb{N}$. Let the interpretations of the function symbols S (successor), $+$ (addition), and the constant 0 be the standard operations over natural numbers. Verify that:

   (a) $M \models \forall w \neg(S(w) = 0)$

   (b) $M \not\models \forall x \exists y(x + y = 0)$

4. Let M be a model with the universe $U^M = \mathbb{Q}$ (the set of rational numbers), and let the binary predicate $<$ be interpreted as the standard strict inequality over the rationals. For each of the following formulas φ, determine whether $M \models \varphi$.

   (a) $\forall x \exists y(x < y)$

   (b) $\exists x \forall y(x < y \rightarrow x = y)$

   (c) $\forall x \forall y \forall z(x < y \rightarrow (y < z \rightarrow x < z))$

5. Show using Natural Deduction:

   (a) $\vdash \forall x \forall y(x = y \rightarrow y = x)$ (Symmetry of Equality)

   (b) $\vdash \forall x \forall y \forall z(x = y \rightarrow (y = z \rightarrow x = z))$

   (c) $\vdash \forall x \forall y(x = y \rightarrow (P(x) \rightarrow P(y)))$

   (d) $\vdash \forall x P(x, x) \rightarrow \forall x \exists y P(x, y)$

   (e) $\vdash \exists x \varphi \rightarrow \forall x \varphi$ if x does not occur free in φ.

Part 2

6. Write an FO formula that will evaluate to true only over a structure that has at least n elements and at most m elements.

7. Consider the vocabulary $\mathcal{V}$ that has the binary function symbols $+, \times$. Let M be the $\mathcal{V}$-structure having as universe U^M the set $\mathbb{N}$ of natural numbers and which interprets $+, \times$ in the usual way. Construct FO formulae Zero(x), One(x), Even(x), Odd(x) and Prime(x) over $\mathcal{V}$ such that, for every environment α,

   - $M, \alpha \models \text{Zero}(x)$ iff $\alpha(x)$ is zero.
   - $M, \alpha \models \text{One}(x)$ iff $\alpha(x)$ is one.
   - $M, \alpha \models \text{Even}(x)$ iff $\alpha(x)$ is even.
   - $M, \alpha \models \text{Odd}(x)$ iff $\alpha(x)$ is odd.
   - $M, \alpha \models \text{Prime}(x)$ iff $\alpha(x)$ is prime.

   Goldbach's conjecture says that every even integer greater than 2 is the sum of two primes. Whether or not this is true is an open question in number theory. State Goldbach's conjecture as a FO-sentence over $\mathcal{V}$.

8. Are the following claims correct? Justify your answers.

   (a) $\exists x(P(x) \rightarrow \forall y P(y))$ is valid.

   (b) For any formula φ and constant symbol c, if $\varphi[c/x]$ is valid and c does not appear in φ then $\forall x \varphi$ is valid.

   (c) $\forall x(P(x) \vee Q(x)) \rightarrow (\forall x P(x) \vee \forall x Q(x))$ is valid.

   (d) If x does not appear free in ψ, then the formula $\exists x(\varphi \rightarrow \psi)$ is logically equivalent to $(\forall x \varphi) \rightarrow \psi$.

(e) Natural Deduction for First Order Logic will no longer be sound if empty universes are allowed.

9. Skolem Arithmetic is the first-order theory of the natural numbers with multiplication. Formulae in Skolem Arithmetic (SA) use first-order variables $x, y, \dots$ that range over natural numbers, $\mathbb{N} = \{0, 1, 2, \dots\}$ and the ternary multiplication predicate $x * y = z$. This is the only predicate available to you, other than the usual equality predicate ($=$). Moreover, you are free to use constants from $\mathbb{N}$. Here is a well formed sentence in this logic:

$$\forall x \forall y \exists z (x * z = y)$$

Construct formulae in SA which capture the following:

(a) Divide(a, b) : a divides b.

(b) GCD(a, b, d) : d is the GCD (greatest common divisor) of a and b.

(c) LCM(a, b, e) : e is the LCM (least common multiple) of a and b.

(d) Prime(p) : p is a prime number.

(e) Prime-Power(b) : b is a power of some prime.

(f) k-Product(b) : b is the product of k primes (not necessarily distinct), where k is a fixed number.

10. Let $\mathcal{V}$ be a vocabulary consisting of the binary function symbol $+$ and a constant 0. We denote by $x + y$ the function $+(x, y)$. Consider the following sentences:

$$\varphi_1 := \forall x \forall y \forall z [(x + (y + z)) = ((x + y) + z)]$$
$$\varphi_2 := \forall x [(x + 0) = x \wedge (0 + x) = x]$$
$$\varphi_3 := \forall x [\exists y (x + y = 0) \wedge \exists z (z + x = 0)]$$

Let ψ be the conjunction of the three sentences.

(a) Prove ψ is satisfiable but not valid.

(b) Show that ψ is not equivalent to any of $\varphi_1 \wedge \varphi_2$, $\varphi_2 \wedge \varphi_3$ and $\varphi_1 \wedge \varphi_3$.

(c) Either prove the following formulas are logical consequences of ψ using natural deduction or disprove them via a countermodel:

- $\forall x \forall y ((x + y) = (y + x))$
- $\forall x \forall y \forall z (((x + y = 0) \wedge (x + z = 0)) \rightarrow (y = z))$
- $\forall x \forall y \forall z ((x + y = x + z) \rightarrow (y = z))$
- $\forall x ((x + x = 0) \rightarrow (x = 0))$